

Assignment - 1

MVC Design Pattern Explanation:-

The MVC design pattern is a software architectural pattern used to separate an application into three main components: Model, View & Controller. The separation helps ~~code~~ code for better maintainability, Scalability & testing.

1) Model:

- i) Represents the data & business logic.
- ii) manages data, responds to requests for information & updates the view where the data changes.

2) View:

- i) Represents the user interface.
- ii) Displays data from the model & send the user input to controller.

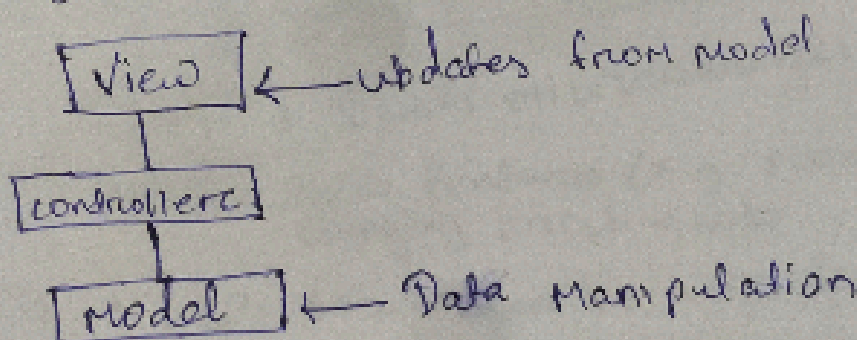
3) Controller

- i) Acts as an intermediary the model & view.
- ii) Takes input from the user, process it & updates the view.

MVC Varieties

- 1) classic MVC:- In this variant, the view directly interacts with the models the controller process input, updates the model & notifies the view of changes.

Diagram



24 Model View Presenter

Here the presenter replaces the controller & takes over the responsibility of updating both the model & the view. The view is passive & relies on the presenter for all interactions.

When to use:

Best for applications with complex UI interaction like desktop, where the view is decoupled from the logic.

Diagram:

